

# Gaussian $\alpha$ -projections are not coordinatewise variance-monotone

An exact counterexample to AIM problem 558,  
with a determinant-monotonicity theorem

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## Abstract

For a fixed correlated Gaussian target, consider its optimal diagonal Gaussian approximation under  $D_\alpha(p||q)$ , with  $\alpha > 1$ . We give a rational three-dimensional target whose optimal covariance is the identity at  $\alpha = 10$ , but whose second and third variances are strictly smaller than one for every  $\alpha > 10$ . The first variance is strictly larger than one, so the covariances are incomparable in the coordinatewise order. A convex comparison argument proves the result without numerical optimization or limiting arguments. This disproves the statement in AIM problem 558 and the corresponding conjecture in Remark 12 of Margossian, Pillaud-Vivien and Saul. We also derive a general sensitivity formula, show that dimension three is minimal, and prove that the determinant of the optimal covariance nevertheless increases strictly for every non-diagonal Gaussian target and every  $\alpha > 1$ .

**Evidence status.** Complete proof claim, self-audited with exact rational and symbolic calculations. No independent human review, publication, or formal proof-assistant verification is claimed. The appropriate repository status pending independent review is *Solution claimed*.

## 1 The target and the result

Let  $p = N(\mu, \Sigma)$ , where  $\Sigma \succ 0$  is real, symmetric and non-diagonal. Optimize over  $q = N(\nu, \Psi)$  with positive diagonal  $\Psi$  the divergence

$$D_\alpha(p||q) = \frac{\int_{\mathbb{R}^d} p(x)^\alpha q(x)^{1-\alpha} dx - 1}{\alpha(\alpha - 1)}, \quad \alpha > 1. \quad (1)$$

The target in AIM problem 558 is coordinatewise monotonicity of the separately optimized variances as  $\alpha$  increases [2]. The ordering is the one in Definition 2 of [1]; its extension beyond one is conjectured in Remark 12, p. 26. The proof below addresses that ordering, not an ordering of divergence values for fixed distributions.

**Theorem 1** (Exact counterexample). *Set  $\mu = 0$  and take the target precision*

$$K = \Sigma^{-1} = \begin{pmatrix} 37/5 & -4 & -4 \\ -4 & 243/70 & 17/7 \\ -4 & 17/7 & 243/70 \end{pmatrix}. \quad (2)$$

*This matrix is positive definite. The unique optimal diagonal covariance satisfies*

$$\Psi(10) = I_3, \quad \Psi_{11}(\alpha) > 1, \quad \Psi_{22}(\alpha) = \Psi_{33}(\alpha) < 1 \quad (\alpha > 10). \quad (3)$$

*In particular,  $(\alpha_1, \alpha_2) = (10, 11)$  disproves the conjecture.*

For completeness, the target covariance itself is

$$\Sigma = \begin{pmatrix} 295/583 & 200/583 & 200/583 \\ 200/583 & 33910/42559 & -6900/42559 \\ 200/583 & -6900/42559 & 33910/42559 \end{pmatrix}. \quad (4)$$

The result concerns uncertainty reported by an exactly optimized factorized approximation. It does not claim that an arbitrary non-Gaussian posterior is calibrated, nor does it contradict the ordering established for  $0 < \alpha < 1$  in [1].

## 2 A convex formulation and global optimality

Write  $a = \alpha > 1$ ,  $b = a - 1$ ,  $t_i = 1/\Psi_{ii}$ , and  $T = \text{diag}(t)$ . Translation allows us to assume  $\mu = 0$ . Define

$$H_a(t) = aK - bT, \quad S_a(t) = H_a(t)^{-1}, \quad (5)$$

$$\mathcal{D}_a = \{t \in (0, \infty)^d : H_a(t) \succ 0\},$$

$$F_a(t) = -\log \det H_a(t) - b \sum_{i=1}^d \log t_i. \quad (6)$$

If  $H_a(t)$  is not positive definite, the Gaussian integral in (1) diverges. When  $H_a(t) \succ 0$ , completing the square gives

$$\begin{aligned} 2 \log \int p^a q^{1-a} &= -a \log \det \Sigma + F_a(t) \\ &\quad + b \nu^T T \nu + b^2 \nu^T T S_a(t) T \nu. \end{aligned} \quad (7)$$

The last two terms are minimized uniquely at  $\nu = 0$ . Because  $a(a-1) > 0$ , minimizing the divergence is therefore equivalent to minimizing  $F_a$  on  $\mathcal{D}_a$ .

**Proposition 2** (Existence, uniqueness and stationarity). *For every  $a > 1$  and  $K \succ 0$ ,  $F_a$  has a unique global minimizer  $t^*(a)$  in  $\mathcal{D}_a$ . It is characterized by*

$$[S_a(t^*(a))]_{ii} = \frac{1}{t_i^*(a)} \quad (1 \leq i \leq d). \quad (8)$$

*The minimizer is smooth in  $a$  and in  $K$  locally on their domains.*

*Proof.* The domain is nonempty (take all  $t_i$  sufficiently small) and convex. Differentiation gives

$$\partial_i F_a(t) = b \left( [S_a(t)]_{ii} - \frac{1}{t_i} \right), \quad (9)$$

$$\nabla^2 F_a(t) = b^2 (S_a(t) \circ S_a(t)) + b \text{diag}(t_i^{-2}) \succ 0, \quad (10)$$

where  $\circ$  is the entrywise product. Indeed, for  $S \succ 0$  and real  $w$ ,

$$w^T (S \circ S) w = \text{tr} \left[ (S^{1/2} \text{diag}(w) S^{1/2})^2 \right] \geq 0,$$

and the second term in (10) is positive definite.

The domain is bounded, since  $0 < t_i < aK_{ii}/b$ . As a sequence approaches its boundary, at least one  $t_i$  tends to zero or  $H_a(t)$  becomes singular. The corresponding negative logarithm diverges to  $+\infty$ ; the remaining terms are bounded below because both  $t$  and  $H_a(t)$  are bounded above. Thus a minimizer exists in the interior. Strict convexity proves uniqueness and makes (8) sufficient as well as necessary. The implicit function theorem applied to (9), with invertible Hessian, proves local smoothness.  $\square$

### 3 Proof of the counterexample

#### 3.1 An exactly known optimum at $a = 10$

Introduce the correlation matrix

$$R = \begin{pmatrix} 1 & 4/5 & 4/5 \\ 4/5 & 1 & 3/10 \\ 4/5 & 3/10 & 1 \end{pmatrix}. \quad (11)$$

Its leading principal minors are 1,  $9/25$ , and  $7/500$ , so  $R \succ 0$ . Direct rational calculation gives

$$R^{-1} = \begin{pmatrix} 65 & -40 & -40 \\ -40 & 180/7 & 170/7 \\ -40 & 170/7 & 180/7 \end{pmatrix}, \quad K = \frac{R^{-1} + 9I}{10}. \quad (12)$$

Consequently  $K \succ 0$ , and  $H_{10}(\mathbf{1}) = R^{-1} \succ 0$ . Moreover,

$$S_{10}(\mathbf{1}) = R, \quad [S_{10}(\mathbf{1})]_{ii} = 1.$$

By Proposition 2,  $t^*(10) = \mathbf{1}$  and  $\Psi(10) = I_3$  exactly.

#### 3.2 A feasible rational comparison curve

For  $a > 10$ , put

$$c(a) = \frac{213a + 370}{5(49a + 10)}, \quad \bar{t} = (c(a), 1, 1). \quad (13)$$

Then  $0 < c(a) < 1$ , because

$$1 - c(a) = \frac{32(a - 10)}{5(49a + 10)} > 0.$$

In the orthonormal basis  $e_1, (e_2 + e_3)/\sqrt{2}, (e_2 - e_3)/\sqrt{2}$ , the matrix  $H_a(\bar{t})$  is

$$\begin{pmatrix} 37a/5 - bc(a) & -4\sqrt{2}a & 0 \\ -4\sqrt{2}a & h_+ & 0 \\ 0 & 0 & h_- \end{pmatrix}, \quad h_+ = \frac{49a + 10}{10}, \quad h_- = \frac{3a + 70}{70}. \quad (14)$$

Both  $h_+$  and  $h_-$  are positive. The Schur complement of  $h_+$  in the  $2 \times 2$  block equals  $c(a)$ , since

$$c(a) = \frac{37}{5} - \frac{32a}{h_+}, \quad \frac{37a}{5} - bc(a) - \frac{32a^2}{h_+} = c(a). \quad (15)$$

Thus  $\bar{t} \in \mathcal{D}_a$ , and block inversion gives

$$[S_a(\bar{t})]_{11} = \frac{1}{c(a)}, \quad (16)$$

$$[S_a(\bar{t})]_{22} = [S_a(\bar{t})]_{33} = \frac{1}{2} \left( \frac{1}{h_-} + \frac{1}{h_+} + \frac{32a^2}{c(a)h_+^2} \right). \quad (17)$$

Substitution and reduction to a common denominator yield the exact identity

$$\boxed{[S_a(\bar{t})]_{22} - 1 = -\frac{a(a-10)(7311a-64010)}{(3a+70)(49a+10)(213a+370)} < 0 \quad (a > 10).} \quad (18)$$

The last sign follows from  $7311a - 64010 > 9100$  for  $a > 10$ . It follows from (9) that

$$\nabla F_a(\bar{t}) = (0, g, g)^T \quad \text{for some } g < 0. \quad (19)$$

### 3.3 From the comparison point to the global optimum

Let  $t^* = t^*(a)$ . Interchanging coordinates 2 and 3 leaves  $K$ ,  $F_a$ , and  $\mathcal{D}_a$  unchanged. Uniqueness therefore implies  $t_2^* = t_3^*$ .

The comparison point is not stationary, so  $F_a(t^*) < F_a(\bar{t})$ . The supporting-plane inequality for the differentiable convex function  $F_a$  gives

$$F_a(t^*) \geq F_a(\bar{t}) + \nabla F_a(\bar{t})^T (t^* - \bar{t}) = F_a(\bar{t}) + g(t_2^* + t_3^* - 2).$$

Combining these inequalities shows  $g(t_2^* + t_3^* - 2) < 0$ . Since  $g < 0$  and  $t_2^* = t_3^*$ , both leaf precisions are greater than one. Hence

$$\Psi_{22}(a) = \Psi_{33}(a) < 1 \quad (a > 10). \quad (20)$$

For the first coordinate, set  $u = t_2^* = t_3^* > 1$ . Applying the first stationarity equation to the same block decomposition now at  $t^*$  gives

$$t_1^* = \frac{37}{5} - \frac{32a}{59a/10 - bu}.$$

Positive definiteness implies  $59a/10 - bu > 0$ . Because  $u > 1$ , this denominator is less than  $h_+$ , so

$$0 < t_1^* < \frac{37}{5} - \frac{32a}{h_+} = c(a) < 1.$$

Thus  $\Psi_{11}(a) > 1$ . This finishes the proof of Theorem 1.

### 3.4 A single finite rational certificate: $a = 11$

Only the finite pair (10, 11) is needed to disprove the conjecture. At  $a = 11$ , the preceding comparison point and inverse become

$$\bar{t} = \left( \frac{2713}{2745}, 1, 1 \right), \quad S_{11}(\bar{t}) = \begin{pmatrix} \frac{2745}{2713} & \frac{2200}{2713} & \frac{2200}{2713} \\ \frac{2200}{2713} & \frac{153231490}{153412011} & \frac{48970900}{153412011} \\ \frac{2200}{2713} & \frac{48970900}{153412011} & \frac{153231490}{153412011} \end{pmatrix}. \quad (21)$$

Here  $\det H_{11}(\bar{t}) = 279439/3500 > 0$ , and

$$\nabla F_{11}(\bar{t}) = \left( 0, -\frac{1805210}{153412011}, -\frac{1805210}{153412011} \right)^T.$$

Feasibility follows from (14)–(15), or from the exact positive leading principal minors checked by the accompanying program. The same supporting-plane argument proves the contradiction using rational data alone.

Numerical values are supplied only to indicate scale:

$$\Psi(11) \approx \text{diag}(1.012120833804016, 0.999727697047115, 0.999727697047115). \quad (22)$$

As an additional algebraic cross-check, putting  $y = \Psi_{22}(11) = \Psi_{33}(11)$  gives

$$\Psi_{11}(11) = \frac{5(649y - 100)}{6413y - 3700},$$

and the feasible solution near one satisfies

$$303828701y^3 - 573214290y^2 + 295302000y - 25900000 = 0.$$

The polynomial is negative at  $9997/10000$  and positive at  $4999/5000$ . Neither this polynomial nor the decimal approximation is needed for the proof.

## 4 A general sensitivity identity

The counterexample can also be seen in an exact negative derivative. The same calculation leads to positive structural results.

**Proposition 3** (Sensitivity formula). *Let  $v_i(a) = \Psi_{ii}(a)$ ,  $D = \text{diag}(v_i)$ , and*

$$S = (aK - (a - 1)D^{-1})^{-1}, \quad R_a = D^{-1/2}SD^{-1/2}, \quad C = R_a \circ R_a.$$

*Then  $R_a$  is a positive-definite correlation matrix. The logarithmic variance derivatives  $u_i = v'_i/v_i$  obey*

$$\boxed{[I + (a - 1)C]u = \frac{C\mathbf{1} - \mathbf{1}}{a}.} \quad (23)$$

*Proof.* Stationarity implies  $S_{ii} = v_i$ , so  $R_a$  has diagonal entries one. Put  $T = D^{-1}$  and  $U = \text{diag}(u)$ . Since  $T' = -TU$ , differentiation of  $S^{-1} = aK - (a - 1)T$  gives

$$S' = -S(K - T + (a - 1)TU)S.$$

Use  $K - T = (S^{-1} - T)/a$  to rewrite this as

$$S' = \frac{STS - S}{a} - (a - 1)STUS.$$

Now  $(STS)_{ii}/v_i = \sum_j C_{ij}$  and  $(STUS)_{ii}/v_i = \sum_j C_{ij}u_j$ . Taking diagonal entries and dividing by  $v_i$  proves (23).  $\square$

For the target (2),  $D(10) = I$  and  $R_{10} = R$  from (11). Exact solution of (23) yields

$$\left. \frac{d}{da} \text{diag } \Psi(a) \right|_{a=10} = \left( \frac{3392}{260905}, -\frac{91}{521810}, -\frac{91}{521810} \right)^T. \quad (24)$$

The right side of (23) is entrywise nonnegative, but the inverse of its positive-definite coefficient matrix need not preserve entrywise positivity. The exact derivative displays the resulting failure.

## 5 What remains monotone

### 5.1 Dimension three is minimal

**Corollary 4.** *For a non-diagonal two-dimensional Gaussian target, both optimal variances are strictly increasing throughout  $a > 1$ . Therefore the dimension of Theorem 1 is minimal.*

*Proof.* In two dimensions write

$$R_a = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \quad |r| < 1.$$

Non-diagonality of  $K$  implies  $r \neq 0$ : if  $r = 0$ , then  $S$  and  $S^{-1}$  are diagonal, forcing  $K$  to be diagonal. The two row sums of  $C$  are equal, so (23) gives

$$u_1 = u_2 = \frac{r^2}{a[a + (a-1)r^2]} > 0.$$

Both variances have positive derivatives everywhere. In dimension one the optimal approximation is the target itself, independent of  $a$ .  $\square$

The counterexample extends to every  $d \geq 3$  by adjoining an independent diagonal Gaussian block. The objective splits over the blocks, leaving the three-dimensional optimum unchanged.

### 5.2 Strict increase of generalized variance and entropy

**Theorem 5** (Determinant monotonicity). *For every real non-diagonal positive-definite target covariance and every  $a > 1$ ,*

$$\frac{d}{da} \log \det \Psi(a) > 0. \quad (25)$$

*Thus generalized variance and Gaussian differential entropy increase strictly, even though individual marginal variances can decrease.*

*Proof.* We first show that for any positive-definite correlation matrix  $R \neq I$  in dimension  $d$ ,

$$R \circ R - \frac{1}{d} \mathbf{1}\mathbf{1}^T \succ 0. \quad (26)$$

For a real vector  $w$ , put  $W = \text{diag}(w)$  and  $M = R^{1/2}WR^{1/2}$ . The trace Cauchy–Schwarz inequality yields

$$w^T(R \circ R)w = \text{tr}(M^2) \geq \frac{\text{tr}(M)^2}{d} = \frac{(\mathbf{1}^T w)^2}{d}.$$

Equality requires  $M = \lambda I$ , equivalently  $W = \lambda R^{-1}$ . If  $\lambda = 0$ , then  $w = 0$ . If  $\lambda \neq 0$ , diagonality of  $W$  forces  $R^{-1}$ , hence  $R$ , to be diagonal. A diagonal correlation matrix is  $I$ , contrary to assumption. Therefore equality is impossible for nonzero  $w$ , proving (26).

Apply this to  $R_a$  in Proposition 3. It is not  $I$ , since that would force  $K$  to be diagonal. With  $b = a - 1 > 0$  and  $B = I + bC$ , we obtain

$$B \succ I + \frac{b}{d} \mathbf{1}\mathbf{1}^T, \quad \text{hence} \quad \mathbf{1}^T B^{-1} \mathbf{1} < \mathbf{1}^T \left( I + \frac{b}{d} \mathbf{1}\mathbf{1}^T \right)^{-1} \mathbf{1} = \frac{d}{a}.$$

Equation (23) can be written

$$u = \frac{1}{ab}(\mathbf{1} - aB^{-1}\mathbf{1}).$$

Summing its entries gives

$$\frac{d}{da} \log \det \Psi(a) = \mathbf{1}^T u = \frac{d - a\mathbf{1}^T B^{-1}\mathbf{1}}{ab} > 0.$$

The entropy conclusion follows from  $h(N(\mu, \Psi)) = \frac{d}{2} \log(2\pi e) + \frac{1}{2} \log \det \Psi$ . □

## 6 Reproduction, provenance and scope of the claim

The accompanying `verify_558.py` uses only Python’s standard-library rational arithmetic. It checks positive definiteness by principal minors, the exact target inverse, stationarity at  $a = 10$ , the feasible comparison point and negative gradient at  $a = 11$ , and the derivative (24). Run

```
python verify_558.py
```

The optional `symbolic_audit_558.py` verifies the rational-function identity (18) using SymPy, then separately solves the original stationarity equations with mpmath at 90-digit precision. The script requires positive precision coordinates, a positive-definite  $H$ , a maximum stationarity residual below  $10^{-75}$ , and the stated direction of the variance change. The accompanying log records the measured values. These numerical checks are not used in the proof.

**Provenance.** On 1 October 2026, problem 558 was rechecked at MColbrook/AIM revision `aa776a01d7d48a79f93251af11fde9454b0aea95`; see the pinned target link in [2]. It was marked Open, last checked 24 September 2026. The supplied package records direct checks of Definition 2 and Remark 12 of the published source.

**Review.** The original proof and audits were AI-generated. Codex (AI assistant) subsequently read the argument, checked the pinned target and reran the audits. Contributor and proposed submitter: Siavash Sadeghi. No independent human review, publication priority or formal verification is claimed. The status remains *Solution claimed* [3]. The accompanying report maps the target hypotheses to the proof. No external submission was made.

## References

- [1] C. C. Margossian, L. Pillaud-Vivien and L. K. Saul. *Variational Inference for Uncertainty Quantification: an Analysis of Trade-offs*. Journal of Machine Learning Research **26**(202) (2025), 1–41. Definition 2, Theorem 3 and Remark 12. <https://www.jmlr.org/papers/v26/24-0878.html>. Also arXiv:2403.13748v5, 19 October 2025.
- [2] MColbrook/AIM. *Problem 558: Variance ordering for Gaussian alpha-divergence approximations*. Repository revision `aa776a01d7d48a79f93251af11fde9454b0aea95`, accessed 1 October 2026. [https://github.com/MColbrook/AIM/blob/aa776a01d7d48a79f93251af11fde9454b0aea95/problem s/558-gaussian-variational-variance-ordering.md](https://github.com/MColbrook/AIM/blob/aa776a01d7d48a79f93251af11fde9454b0aea95/problem%2F558-gaussian-variational-variance-ordering.md).
- [3] MColbrook/AIM. *Contributing: Reporting a resolution; Status and evidence*. Accessed 1 October 2026. <https://github.com/MColbrook/AIM/blob/aa776a01d7d48a79f93251af11fde9454b0aea95/CONTRIBUTING.md>.